



University of Brasilia

Institute of Exact Sciences
Computer Science Department

Slot Allocation Problem: Medium hub airport

Lucas Kuniyoshi

Dissertation presented as a partial requirement for the qualification of the
Professional Master's Degree in Applied Computing

Advisor

Prof. Dr. André Luiz Peron Martins Lanna

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Prof. Dr. André Luiz Peron Martins Lanna (Advisor)
FGA/UnB

Prof. Dr. XXXX Dr. YYYY
Stanford University Microsoft Research

Prof. Dr. Gladston Luiz da Silva
Coordinator of the Graduate Program in Applied Computing

Brasilia, March 20th, 20XX

Resumo

A alocação de slots das companhias aéreas dentro da infraestrutura aeroportuária influencia diretamente a lucratividade de um aeroporto e sua eficiência em absorver o crescente tráfego de transporte aéreo de passageiros (ACI World Insights, 2025). Embora a expansão da capacidade aeroportuária represente a solução definitiva para o desequilíbrio entre esse crescimento e a infraestrutura limitada (IATA, 2025), fatores como regulamentações, áreas delimitadas e limitações financeiras frequentemente exigem que os aeroportos implementem mecanismos alternativos de gestão da demanda. Esta tese tem como objetivo apresentar um método para resolver o problema de alocação de slots aeroportuários, focando em um modelo que emprega Programação Inteira (IP), Aprendizado de Máquinas (ML) e Teoria das Filas (QT). Especificamente, QT é utilizada para simular a capacidade real de um aeroporto, identificando gargalos. ML, por sua vez, busca preencher lacunas de informação cruciais para um processo de alocação aprimorado, particularmente detalhes frequentemente retidos pelas companhias aéreas para projeções de longo prazo. IP tem sido amplamente adotada na literatura como um mecanismo de gestão de slots, levando à redução de atrasos e deslocamento de slots para as companhias aéreas. Adicionalmente, uma Literatura Sistemática de Literatura (SLR) foi conduzida, investigando as abordagens existentes para o problema de alocação de slots. Este processo gerou um protocolo de revisão, dados de pesquisa e um relatório abrangente detalhando o conhecimento avaliado sob múltiplas perspectivas, as metodologias empregadas e as áreas identificadas para pesquisa futura. Consequentemente, esta tese oferece uma contribuição tríplice, demonstrando uma aplicação unificada desses métodos para abordar o problema: (i) Uma SLR abrangente do problema de alocação de slots, incluindo um protocolo que garante a reprodutibilidade e validade da revisão, juntamente com uma síntese narrativa explorando as principais motivações, métodos, resultados e limitações do tópico por meio de mapeamento conceitual; (ii) um estudo de caso do Aeroporto Internacional de Brasília (BSB), que utiliza QT para avaliar a capacidade do aeroporto; (iii) uma alocação de submissões iniciais de slots utilizando IP, cujo objetivo é minimizar o deslocamento de slots solicitados pelas companhias aéreas, com informações faltantes obtidas por meio da aplicação de ML.

Palavras-chave: Problema de alocação de slots em aeroportos, Revisão sistemática da literatura, Programação inteira, Aprendizado de máquina, Teoria das filas, e Otimização

Abstract

The allocation of airline flight schedules within airport infrastructure directly influences an airport's profitability and its efficiency in absorbing growing passenger air transportation traffic (ACI World Insights, 2025). While expanding airport capacity represents the ultimate solution to the imbalance between this growth and the limited infrastructure (IATA, 2025), factors such as regulations, delimited areas, and financial limitations frequently require airports to implement alternative demand management mechanisms. This thesis aims to present a method for solving the airport slot allocation problem, focusing on a model employing Integer programming (IP), Machine Learning (ML), and Queueing Theory (QT). Specifically, QT is utilized to simulate an airport's actual capacity, identifying bottlenecks. ML, conversely, seeks to bridge information gaps crucial for an improved allocation process, particularly details often withheld by airlines for long-term projections. IP has been widely adopted in literature as a slot management mechanism, leading to reduced delays and slot displacement for airlines. Additionally, a Systematic Literature Review (SLR) was conducted, investigating existing approaches to the slot allocation problem. This process generated a review protocol, research data, and a comprehensive report detailing the assessed knowledge from multiple perspectives, the methodologies employed and identified areas for future research. Accordingly, this thesis offers a three-fold contribution, demonstrating a unified application of these methods to address the problem: (i) A comprehensive SLR of the slot allocation problem, this includes a protocol ensuring the reproducibility and validity of the review, alongside a narrative synthesis exploring the topic's primary motivations, methods, results, and limitations through conceptual mapping; (ii) a case study of Brasilia International Airport (BSB), which utilizes QT to evaluate the airport's capacity, (iii) an allocation of initial slot submissions using IP, his allocation aims to minimize the displacement of slots requested by airlines, with missing information obtained through ML application.

Keywords: Airports' slot allocation problem, Systematic literature review, Integer programming, Machine learning, Queueing theory, and Optimization

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Chapter 1

Introduction

1.1 Context

The air transportation industry has been growing since 1945. It passed through several economic and geopolitical crises, the most recent being the COVID-19 pandemic (ACI, 2025). The latter has significantly impacted the world passenger traffic, reducing the industry by 40%, when comparing 2021 against 2019, before the spread of the virus (KALIĆ et al., 2022). This diminish of the sector cost approximately 324 billion dollars of gross passenger operating revenue for airlines (HASEGAWA, 2022). The COVID-19 also influenced the airport's revenue and operations. At the beginning of the pandemic, the close sky policies allowed only essential flights, reducing the use of the airport's infrastructure. Sanitary measures also changed the aircraft movement itinerary, due to mandatory cleaning and inspections before and after transporting passengers, which directly affected the demand that could be accepted by the airports. Even through all these negative indicators, it is estimated a recovery of almost 103% of worldwide passenger transportation in 2024, when comparing against 2019 (IATA, 2022). Some regions, such as Central America, have projections that surpass the passenger demand of 2019 at the end of 2023. Therefore, even with this uncertainty, the increasing demand required by the airlines, to compose a better network, will result in congestion of several airports, thereat old discussions of how to efficiently resolve the distribution of aircraft operations at airports come back into evidence.

To understand the operational concepts of an airport, it is essential to divide it into two primary components: the airside and the landside (SHERRY, 2009). During an arrival, the airside process begins when air traffic management guides an aircraft to land on the runway and taxi to its designated parking stand. Once parked, passengers disembark and transition through the gates into the landside terminal. Inside the terminal, arriving passengers can utilize commercial facilities before claiming their luggage and exiting via

ground transportation. Conversely, departing passengers begin at the landside by arriving via ground transport, completing check-in and security, and moving through the terminal to embark. Finally, the aircraft returns to the airside by pushing back from the gate, taxiing to the runway, and taking off.

With the prototypical airport conception, it is possible to start the discussion about the slot allocation problem and how it is related to the way an airport generate its revenue and the main limitations related to its infrastructure.

The airport revenue is mainly generated through aeronautical and non-aeronautical sources, both of which are directly related to the aircraft traffic volume. The former is composed by: the fare paid by each passenger using the airport infrastructure, usually included in the flight cost paid to the airlines; and the movement charge, which is paid by the airline when it lands, uses the infrastructure (e.g., hangar and terminal), and takes-off from the airport. The latter is the revenue generated by: the commercial concession at the terminal, such as rental fees for the retail at the airport area; the real estate exploitation, which includes hotels, restaurants, and parking lots at the airport perimeter; and services offered to passengers and airlines. A market equilibrium between the two revenues sources is needed, since reduction of the aeronautical charges commonly means stimulation to the non-aeronautical revenues, through passenger demand increments, but it does not guarantee the a overall maximum return (ICAO, 2013).

In 2019, the european aeronautical sources generated 54% of overall airport revenue (ACI, 2021), with 63% of that aeronautical income originating from charges directly applied to passengers (ICAO, 2013). Furthermore, airport users often do not pay the full cost of the infrastructure they utilize, as evidenced by the fact that 69% of airports worldwide were loss-making in 2012 (ACI, 2021). To maximize profitability, airports have expanded their variety of commercial activities, which cushions the impact of lower passenger and freight volumes due to the higher profit margins associated with these services. This creates a cross-subsidization between aeronautical and non-aeronautical revenues sources, since profits from commercial exploitation, for example, can be reinvested in airport infrastructure, increase the demand of passenger or aircraft that can be accepted at the airport, reducing capital needs and overall cost.

Knowing that airport are important for a country social-economic-growth and, as it was said, with an efficient management it can achieve profitability, governments have decided that, under the right economic conditions, they can included private sector participation (e.g., outright ownership, short or long-term concessions, public-private-partnership schemes, and management contracts) for the financing and operation of an airport infrastructure. The airports, under this new ownership model, have sought for better ways to compete for both passenger and airlines. The aviation community agrees with the

need of infrastructure investments to accommodate the growth of the industry, increasing the profit margins. Since most of the infrastructure added are in large increments, and the time needed for these ventures is extensive, the airports are exposed to considerable risk (ICAO, 2013). Even with the constant pressure for the airport expansion, most of the administrator have to better manage its current infrastructure, until the investment risks are reduced to an acceptable level. When the expansion of infrastructure it is not possible, the optimization of the allocation in the available slots is strongly sought by the airport administrator, to improve the revenue generation prospects. Some regions have already implemented more restrictive on-time performance goals for airlines (RAVIZZA et al., 2014), which aims to enhance aircraft’s taxi times, take-off sequencing, and allocating push-back time, thus affecting the demand of movements that an airport can accept.

The limitation of airport infrastructure use is determined by several interconnected factors, including the number of aircraft movements runways can accept within specific time intervals and the availability and category of aircraft stands (ARC, 2021; IATA, 2020b). Capacity is further defined by the number of gates available for passenger movements, the processing limits of security inspections at check-in, and the total passenger volume the terminal area can support. Additionally, infrastructure use is constrained by the number of aircraft permitted through time slots and specific environmental restrictions. Ultimately, the most restrictive infrastructure capacity – or a combination of these various limitations – is used to manage airport movements and distribute slots effectively without compromising the level of service through delays or taxiway congestion.

1.2 Problem Definition

The present study introduces two main reasons to elevate the efficiency in the initial slot allocation in an airport. The first reason is related to the airport’s ways to generate revenue. The two main ways of an airport generates revenue (i.e., aeronautical and non-aeronautical) are directly related to the passenger traffic volume. Commercial revenues play a vital role in global airport economics, accounting for 52.9% of total revenue in Africa and the Middle East, 52.6% in North America, 45.7% in Asia-Pacific, and 38.1% in Europe, yet they comprise only 29.0% in Latin America and the Caribbean, where a lack of traffic stimulus often results in a heavy dependence on regulated fares and underdevelopment in both commercial and aeronautical sectors (GRAHAM, 2009). In the Economic Commission for America Latin and the Caribbean (PLANZER; PÉREZ, 2019), it is demonstrated the growth of the worldwide air transportation market, before the COVID-19 pandemic (2017), which, by International Civil Aviation Organization (ICAO), would reach 10 billion passengers per year by 2040. Despite the great projec-

tion, the report also describes the Latin American and the Caribbean challenges to build airport infrastructure, the major impediment to the development of the local sector. The International Air Transport Association (IATA) has urged the local aviation authorities to ensure, between several highlighted topics, the efficient use of the available airport's capacity in the region.

The second reason is related to the impact of the operational performance improvement in the airlines. As it was mentioned, the operational capacity of an airport, i.e., the ability to absorb the demand of flights, is directly related to its infrastructure (e.g., number of runways, gates, aircraft stands, and terminal areas). Although the mentioned features of an airport can be used to determine its capacity, a lower limit is set to mitigate the negative impacts of the airport's main partners, the airlines. This measure goes in the opposite direction of the first reason. When the limit is set greater than it should be, operational impacts are raised (e.g., delays). Since most airlines interconnect their flights within a network, a single delay can spread repercussions far beyond the local level, impacting multiple aircraft itineraries and significantly elevating an airline's operational costs. This topic is discussed in (BALL et al., 2010), where it is described that the airline's costs of crew management, disrupted passenger accommodation and aircraft re-position, due to delays, can reach 20 billion dollars a year (2017). Another negative impact is related to intense and unplanned aircraft taxi traffic. Fuel consumed in stop-and-go situations, primarily resulting from congestion on the airport's taxiway system, can account for approximately 18% of the fuel used during the entire operation (NIKOLERIS; GUPTA; KISTLER, 2011).

For that reason, while both airports and air operators want to expand their operations, a balance between accepted demand, which increases revenues (for airports), and operational efficiency, which mitigates high operational costs (for airlines), is needed.

1.3 Justificative

1.4 Objectives

1.5 Text Structure

Chapter 2

State of the art

2.1 Review planning

2.2 Review execution

2.3 Review results

2.4 Conclusions

Chapter 3

Theoretical Background

3.1 Airport regulations and operation

In order to better understand this study's main problem, it is important to have knowledge of the basic concepts of airport's operation. The prototypical airport is composed of 2 parts: an (i) airside and (ii) landside (SHERRY, 2009). From an operational airport's perspective, the aircraft movement starts at the airside, when it approaches the runway through the commands of the Air Traffic Management (ATM). After the landing at the runway, the aircraft continues the movement, now at the ground level, through the taxiways. The aircraft approaches the apron in the direction of the parking stand. When it is positioned at its designated stand, the aircraft parks and bridges or airstairs are attached to the equipment in order to disembark the passengers, leading them to a gate. After the gate, the passengers are at the landside, in the airport terminal. There, they may enjoy the commercial stores (e.g., convenience stores, restaurants, and car rental agencies) before collecting their luggage and leaving the airport by ground transport or connecting to other flights.

The opposite movement is also possible: the passengers arrive at the airport using ground transportation, they may also use the airport's parking lots; they access the terminal, after check-in procedures and security inspections; they may enjoy the commercial stores, before embarking to the aircraft; at the airside, the aircraft pushes back, after it gets disattached from the airport bridges or airstairs; it proceeds through the taxiways until it arrives at the runways; and finally it takes-off, ending its movement at the airport.

As it is highlighted in (CAVUSOGLU; MACARIO, 2021), there are three predominant processes for the airports' capacity allocation: (i) administrative management, with slot distribution processes through a set of specific prioritization rules; (ii) economic management, where the airport's congestion price is established, slot sales and exchanges are allowed, or auctions are made; and (iii) no regulatory mechanism, relying only on some

infrastructure constraints as parameters (e.g., LaGuardia Airport and Ronald Reagan Washington National Airport as presented by (FAA, 2000)).

The administrative management is the most used mechanism for slots. It is often use the Worldwide Airport Slot Guidelines (WASG), published by IATA(RIBEIRO et al., 2018). There are also applications of WASG variations through local resolutions, as is the case of Brazil, where the local regulator is theNational Civil Aviation Agency of Brazil (ANAC). Therefore, this chapter will: (i) present administrative mechanisms, through the WASG and its allocation practices in congested airports, discussing the differences and similarities between the resolution covering Brazilian airports and the best practices developed by IATA; (ii) briefly explain the characteristics of the economic management, which contains market-driven mechanisms; and (iii) review advantages of slot management mechanism applications compared to airports with almost no restrictions.

3.1.1 Administrative demand management

Demand management has been demonstrated in several works as one of the main processes to distribute air operators' movements in the airports' available infrastructure, dealing with the lack of capacity, delays and operational service level decaying due to congestion (ZOGRAFOS; MADAS; ANDROUTSOPOULOS, 2017; RIBEIRO et al., 2018).

The administrative demand management approach aims to increase airport efficiency with allocation prioritization. Based upon optimization techniques, such practices have as objective to allocate in a certain number of available resources (airport slots), the dependent activities, i.e., air operators' arrivals and departures movements. The allocation, made through the distribution of slots, must satisfy specific conditions (explained in ??) (ZOGRAFOS; MADAS; ANDROUTSOPOULOS, 2017).

At the strategic level, the administrative management develops a cooperation practice where actions are coordinated between stakeholders, with bilateral communications and frequent reviews, so that the allocation may take place in accordance with all parties. The general objective is to match the requests made by the air operators with the declared infrastructure capacity, maximizing the number of accepted movements and, at the same time, minimizing the air operators' operational cost to when they need to reschedule its flights outside the planned hours, due to the restrictions imposed at the airport (GILLEN; JACQUILLAT; ODONI, 2016). As it was explained, the main administrative management strategy is the IATA's WASG, which has been adopted by local authorities in several countries (ZOGRAFOS; MADAS; ANDROUTSOPOULOS, 2017; FAIRBROTHER; ZOGRAFOS, 2018; CAVUSOGLU; MACARIO, 2021).

In order to better understand the the slot management process, Table 3.1 shows a glossary containing the main terms used by WASG.

Table 3.1: Glossary of relevant terms in WASG’s slot management. Adapted from (IATA, 2020b).

#	Term	Definition
1	Slots	A permission that is given by a coordinator for a planned operation to use an airport’s infrastructure at a specific date and time.
2	Airport Level	There are three airport levels: (1) for those whose demand does not exceed the infrastructure capacity; (2) for those whose demand causes certain congestion at specific times (e.g., hours, days, days of week), but which can be resolved through mutual agreements between the facilitator and air operators; and (3) for those whose demand exceeds the airport capacity limits, requiring a coordinator for the slots allocation, using WASG’s best practices and prioritizing allocations for better market competitiveness.
3	Infrastructure capacity and airport parameters	Compiled of the necessary parameters for the coordination of slots. Through these, it is possible to identify the operational capacity for allocation that does not exceed the demand limit, providing an adequate service level (e.g., maximum operations per hour / 30 min. / 15 min. / 5 min. on the runway, number of aircraft parking positions available on the apron, number of available gates, airspace limitations, environmental limitations, etc.).
4	Seasons	There are two specific periods where operations take place. The (1) Summer Season, which starts on the last Sunday of March, and the Winter Season, which starts on the last Sunday in October.
5	Series of slots	A minimum of 5 slots allocated for approximately the same time on the same day of the week through a season.
Continued on next page.		

Table 3.1 - Continued from previous page.

#	Term	Definition
6	Slot pool	All slots that will receive allocation priority at level 3 airports, after the historical slots are properly allocated.
7	Historic Slots	Slots with operating precedence at the allocated airport, acquired by a regularity above 80% in the equivalent previous season.
8	Facilitator	The one responsible for collecting data and adjusting movement at level 2 airports.
9	Coordinator	The one responsible for data collection and coordination of slots at level 3 airports.
10	Activity Calendar	Deadlines and Events that manage the process of coordinating movement and slots for each season. It is established two per year.
11	Previous Season Equivalent Season	Last Season of the same name, i.e., if the current season is a summer season, the previous equivalent season is the prior summer season.
12	New entrants	Air operators that have a small number of operations (less than 7) for each day of a season, or companies that do not operate yet and are requesting operations.
13	Annual movements	Are all movements that have specific times in both Summer and Winter seasons.

Continued on next page.

Table 3.1 - Continued from previous page.

#	Term	Definition
14	Messages	Reference to data shared between coordinator, facilitator and air operators (IATA, 2020a). The messages are usually standardized in a format recognized by the industry (e.g., SSIM), containing critical information about the slot, i.e., operations' start date, operations' end date, air operator IATA's code, equipment type (e.g., Boeing 737 MAX), number of seats offered, type of movement air (e.g., Passenger or Cargo), frequency of operation during the week, hours of operation (slots), among others.

3.1.2 Worldwide Airport Slot Guidelines (WASG) - IATA

To provide a standard for slot management and operations planning, the IATA, together with Airports Council International (ACI) and Worldwide Airport Coordinators Group (WWACG), publishes the WASG (IATA, 2020b).

In general terms, the proposal to coordinate airports through the WASG aims to: (i) improve the connectivity of air services, promoting competitiveness in congested airports; (ii) provide slot schedules that meet demand and are consistent across seasons; (iii) ensure a fair, transparent and non-discriminatory slots allocation; (iv) promote efficient use of airports' infrastructure; (v) guarantee market access to new air operators; (vi) provide a flexible slot management system for the industry, which can be adapt through local regulations and market conditions; and (vii) minimize congestion and delays caused by the allocation of slots with no parameters.

Slot Stakeholders

Slot management impacts several groups interested in the growing demand and the efficiency of airport infrastructure use, including air operators as the origin of movement demand and the airport administrator responsible for balancing slot availability with adequate operational levels. According to the WASG, this ecosystem also involves the authorities responsible for airspace control, the coordinators or facilitators tasked with

managing slots at the airport, and the government bodies responsible for regulating airport use. Figure 3.1 illustrates the relationship of stakeholders, showing the information that is exchanged during the slot management process.

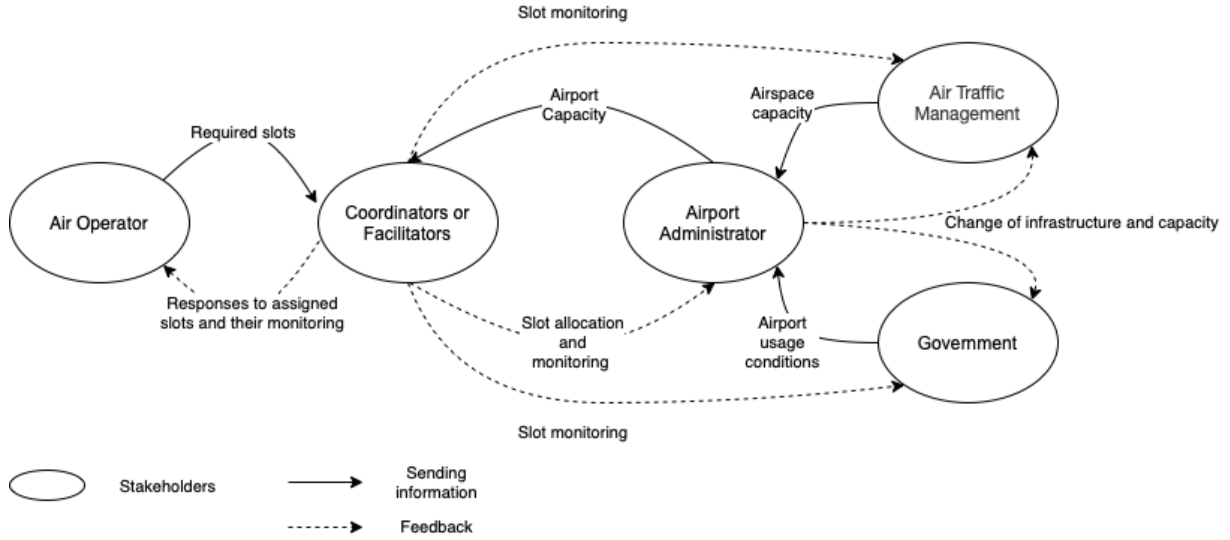


Figure 3.1: Relationship between stakeholders in the management of slots through the direct use of WASG.

Air operators have the responsibility to send requests for desired slots through a message, detailing the operations, and to prepare to cases where congestion is present in the airport and offers with different slots are sent as a response to the request. They also must follow the calendar of coordination activities and local regulations, monitoring the slots allocated to always mitigate the risk of airport infrastructure misuse.

The coordinator or facilitator are responsible to allocate the air operators slots in a transparent and non-discriminatory way. They must ensure that the allocation is made in accordance with the capacity declared by the airport and the priority criteria established in the WASG. During the activities, they assist the understanding of the coordination parameters, local guidelines, regulations, and any other criteria used in the allocation. They must make available the list of the slots allocated and the offers sent, explaining why the requests were not fulfilled. They must attend the conferences, in order to better approach the needs of the air operators, trying to find a middle ground between competitors. They also must perform slot monitoring, to mitigate infrastructure misuse and to apply the use-or-lose-it rules, creating the historical slots.

The airport administrator must provide support to the facilitator or coordinator, assisting in their tasks. Airports must provide the infrastructure information so that capacity is filled with adequate demand. If there is any special limitation present in their

capacity, airports must communicate with the other stakeholders, especially if the demand projection shows to exceed the available capacity in future seasons. Capacity updates must be presented at least twice a year, following each season's activity calendar. The airport must send the monitoring of the slots to the airspace authorities and local regulators.

The authorities responsible for air control must communicate to the airport administrator the airspace capacity that modifies the airport infrastructure capacity parameters. The government that concedes the airport, must, through regulations, send the airport operator the conditions for using the infrastructure, as well as inform the investments that the airport operator needs to do in order to meet the projected target demand.

Capability Analysis

Prior to the slot allocation, a demand and capacity analysis is performed. This analysis must be carried out promptly for the declaration of capacity, one of the season's official activities. It takes into account the airport's ability to absorb movement demand, together with airspace limitations, in a level that the quantity of movements accepted in the airport infrastructure does not degrade its service levels, i.e., does not promote congestion and delays. The result is constraints regarding the runway(s), stand(s), terminal(s), gate(s), among many other relevant airport infrastructures.

Another important limitation of capacity is environmental, i.e., the airport discusses the need to reduce the number of slots available when there is a relevant environmental impact (e.g., periods without movement due to noise pollution or due to environmental events that affects the aircrafts' visual operations).

One of the objectives of the capability statement is to understand which level the airport is at, i.e., Level 1, Level 2, and Level 3. The level indicates the the airport's coordination criticality.

Airport level definition

Each airport level has a different allocation priority. The Figure 3.2 illustrates the allocations by airport level, specifying the prioritized movements.

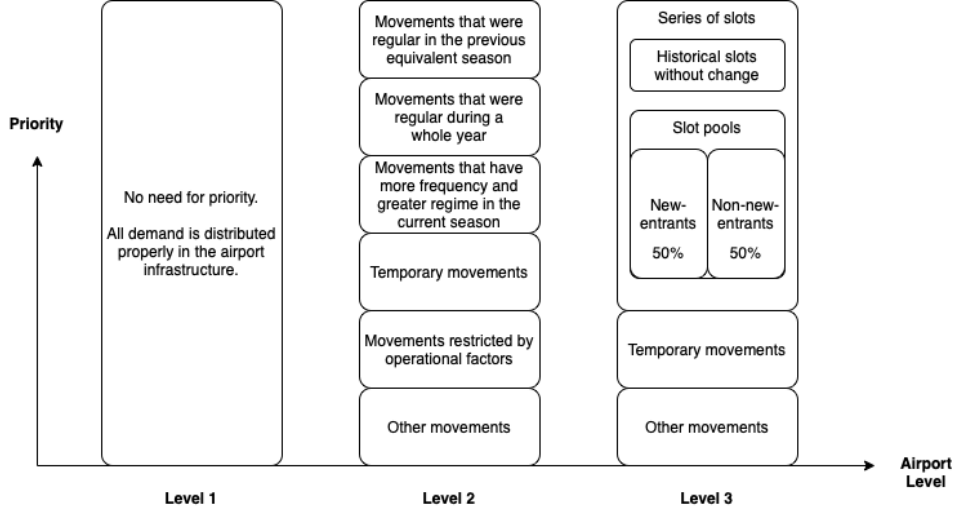


Figure 3.2: Allocation priority for each airport level.

At Level 1, due to the availability of adequate schedules for every demand, the allocation becomes free of priorities. At Level 2, the following allocation priority is applied: (i) movements that operated regularly in the previous equivalent season; (ii) annual movements, i.e., those which have operated with high regularity in the immediately preceding season; (iii) movements with more schedule operations for the current season; (iv) temporary operations; (v) movements restricted by operational factors; and (vi) any other type of movement. At Level 2, as well as a Level 3, a waiting list can be instituted for already congested periods.

For Level 3, WASG proposes a mechanism that grants access to the high-demand market, opening opportunities for new operators to obtain slots at coveted times. For these, the allocation is made prioritizing a series of slots, temporary slots, and other operations. The series of slots can be allocated prioritizing: (i) historical slots with no changes, with changes other than in coordination parameters, or with changes that impact the coordination parameters within a flexibility range period; (ii) slot pool, containing a consistent division of the remaining slots for new entrants and non-new-entrants operators; and (iii) other movements. Additional allocation factors are commonly applied to the Level 3 slot pool based on criteria such as operational frequency, historical performance within flexibility ranges, and regulatory scheduling constraints. Priority may also be determined by waitlist duration, specific service types—such as passenger or cargo—and target markets ranging from regional to international destinations. Furthermore, the selection process incorporates specific parameters established by a stakeholders' committee to ensure optimized infrastructure utilization.

Slot allocation: calendar activities

The slot allocation process follows a structured timeline defined by the WASG activity calendar, beginning with the regulator's publication of the schedule for the upcoming season. Within seven days of this publication, airports must declare their capacity and coordination parameters, after which coordinators provide historic slot details via standard SHL messages. Air operators then review these historic slots through AHD and coordinate with stakeholders to resolve any discrepancies before submitting their initial slot requests (ISD).

Once the initial distribution is communicated via SAL, a dialogue between coordinators and air operators facilitates adjustments and potential slot exchanges during specialized conferences (SC). This is followed by a continuous reallocation phase that lasts until the SRD deadline, at which point all non-operational movements must be canceled. The resulting data forms the reference base (BDR), establishing the 80% regularity threshold required to maintain historic status for future seasons. Finally, the airline community engages in operational planning to ensure that airspace, stands, gates, and other critical infrastructure are prepared to accommodate the scheduled movements.

After all the steps, new slots can be requested if feasible allocation. The season starts with the first planning operation on the last Sunday of October (Winter Season) or on the last Sunday of March (Winter Season). Simultaneously, the planning for the immediately following season takes place. The Figure 3.3 illustrates the season system, with each season indicated by specific colors. Note that during the activities of a specific season, the activities related to the immediately following season are already started, in a continuous process.

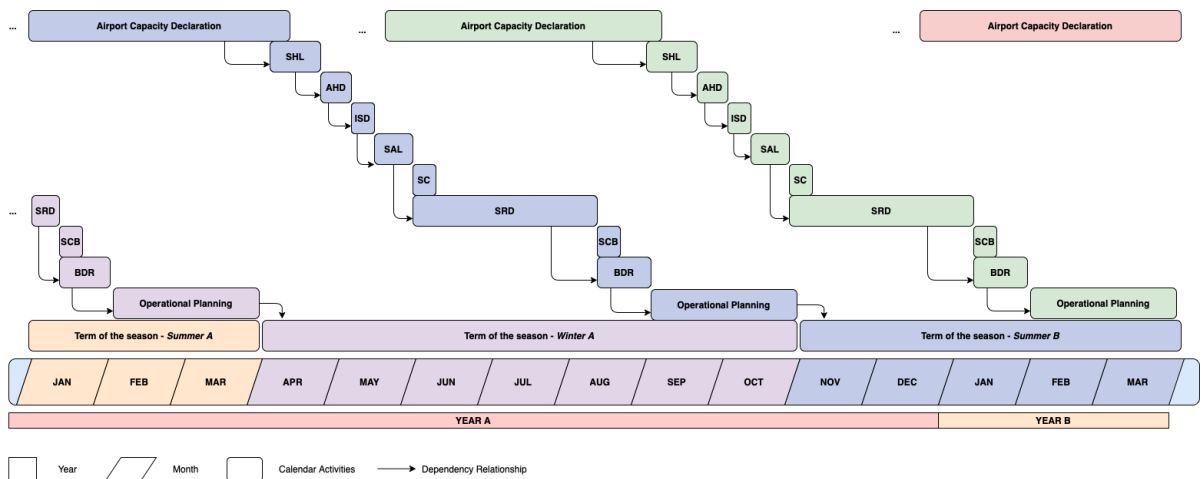


Figure 3.3: Illustration of the calendar of activities stipulated by the WSGA.

Slot Monitoring

For the slots monitoring, WASG recommends that it should be divided into (i) pre-operation analysis, which identifies the slots misuses before the day when the movements occur; and (ii) post-operation analysis, which identifies whether a misuse has occurred and whether the operator has achieved the historic right in its slots. The objective is to ensure proper operation, avoid the waste of airport infrastructure, and open communication channels between stakeholders.

3.1.3 Brazilian local regulations

As recommended by WASG, the application of good practices must consider the regulations in place where the airport is located. Thus, as the present work studies airports in Brazil, it is necessary to understand the rules established by the ANAC, highlighting their similarities and differences.

ANAC establishes, through Resolution No. 682, of July 07th, 2022, the regulations and procedures for allocating slots at airports coordinated and facilitated. As established in WASG, coordinated airports would be Level 3 airports, and airports of interest would be level 2 airports, with their specificities explained in resolution (ANAC, 2022).

The principles of slot coordination are stipulated to minimize the effects of infrastructure congestion at Brazilian airports, focusing on transparency, non-discrimination, impartiality, and efficient use of the declared infrastructure of each airport. It also establishes an activity calendar containing the declaration of capacity, distribution of slot requests, allocation responses, and slots monitoring.

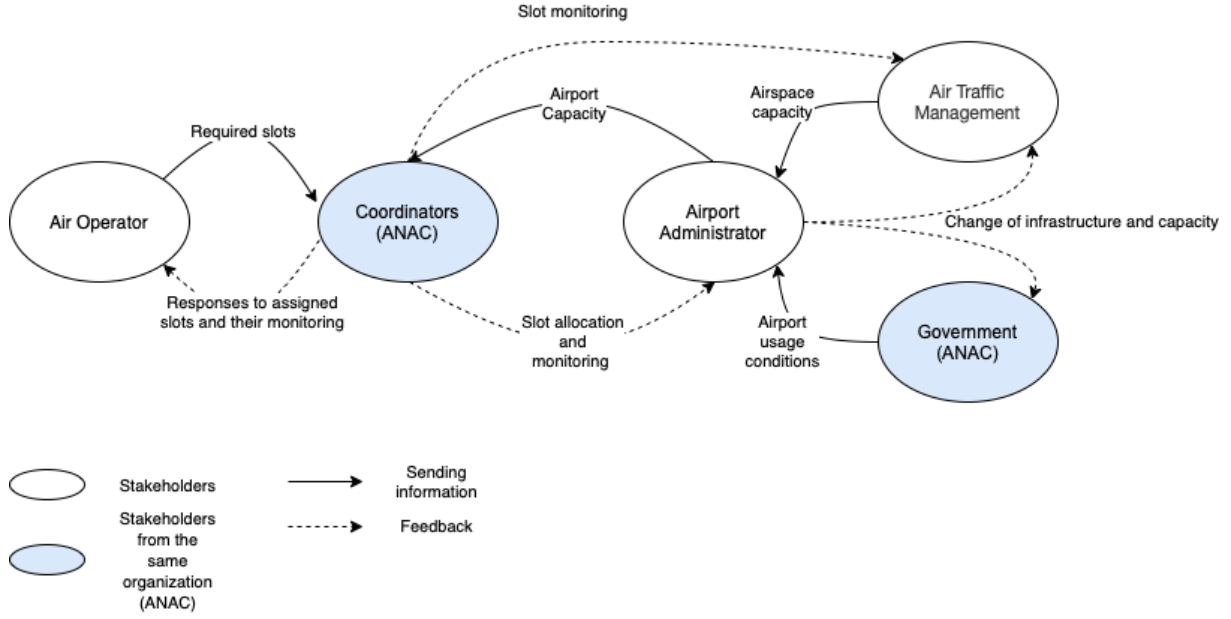


Figure 3.4: Relationship between stakeholders in the management of slots through the use of Brazilian resolution (level 3 airports).

Unlike WASG, the resolution established an ANAC team that will be responsible for activities related to slot coordination at the level 3 airports. The Figure 3.4 illustrates the relationship between stakeholders via ANAC regulation. Thus, if the level of congestion at a given Brazilian airport exceeds the limit of its infrastructure capacity to absorb demand, in any of its allocation parameters (e.g., runway, patio, or terminal), or at due to the request of an interested party, ANAC can declare it as coordinated. In this case, although the agency is also the government agency responsible for monitoring the development of the airport exploration plan, a separate ANAC team will also be in charge of the administrative management of slots.

A significant distinction in the Brazilian slot distribution process is that during the initial submission (ISD), only requests constituting a series of slots are processed, which precludes temporary requests. Furthermore, the priority order differs from WASG standards by prioritizing historic slots first, followed by modified historic slots, and finally new requests. These new operations are the last to be allocated in the SAL, as the system specifically prioritizes the continuity of operations from the immediately preceding season before accommodating new market entries.

Beyond the primary allocation rules, several additional criteria are utilized to resolve ties or conflicts in slot distribution. Priority may be granted based on the total number of operations planned for the season, the use of larger aircraft with higher passenger or cargo capacities, and demonstrated operational or environmental efficiency in previous

equivalent seasons. Furthermore, regulators often favor requests that enhance market competition, specifically those aimed at reducing the Herfindahl-Hirschman Index (HHI) to prevent monopolistic dominance at the airport.

For the analysis of operational efficiency, the resolution presents a comparison of the reference base BDR with the current base of slots for the season, taking into account the operations that did not use the available infrastructure without valid justification. The Regularity Index (IR) is calculated using the Equation 3.1. The IR will be used to determine the historic status of the slots for the next equivalent season.

$$IRs = \frac{Used\ slots}{Slots\ allocated\ in\ the\ BDR} \quad (3.1)$$

ANAC also mitigates the misuse of slots through regulations on infractions and their respective administrative measures. If an air operator obtains a slot that it does not intend to use, wasting the airport infrastructure above the imposed limit, it will be considered an infraction punishable by fines established by the agency, as well as the impediment to obtain historic slots.

Similar to WASG, if the company falls below the minimum regularity target (in this case established by the coordinator), as well as when the intentional misuse of allocated slots is identified, the company may lose the historic slot. In the ANAC's regulation, when misuse happens, if the slots are requested at the next equivalent season, they would be inserted in the slots pool.

In Brazil, it is set to facilitated airports (i.e., level 2 airports) to prioritize the allocation of air operators' slots series that operated in the last season (i.e., the annual movements). Since there are no historic slots, the resolution recommends lower priority for operators who misuse airport infrastructure. The Figure 3.5 illustrates the stakeholders in an facilitated airport configuration stipulated by ANAC. Note that both the airport and the slot facilitator are the same entity.

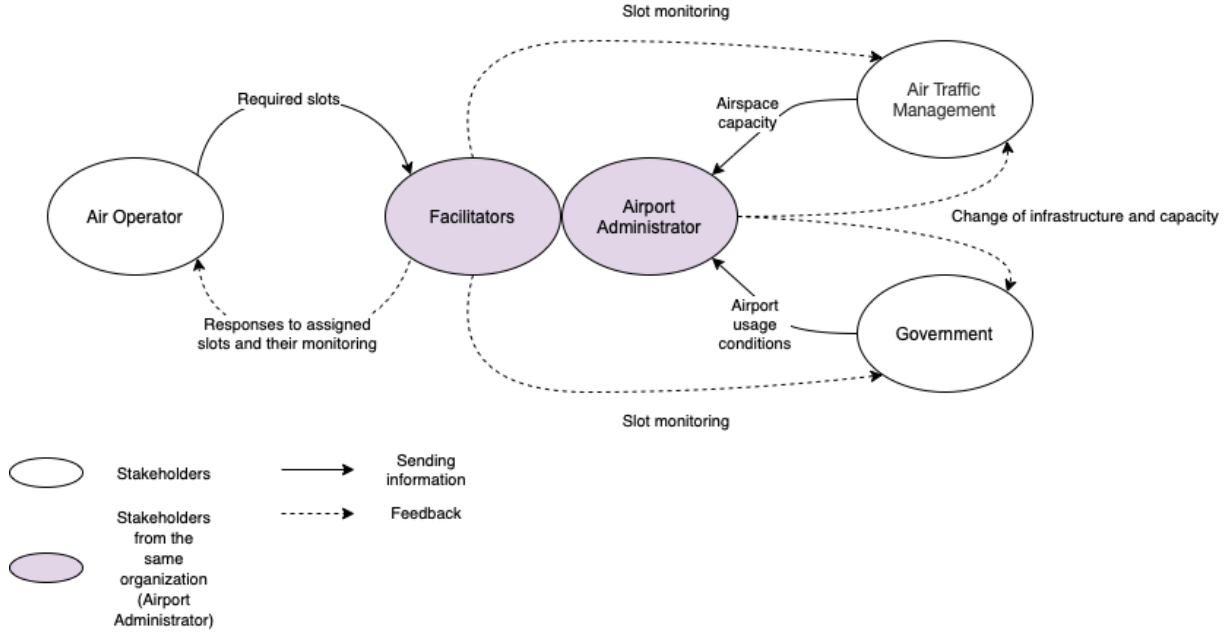


Figure 3.5: Relationship between stakeholders in the management of slots through the use of Brazilian resolution (level 2 airports).

3.1.4 Economic demand management

Although it is not the scope of the present study, research with market-driven approaches, i.e., economic, has been developed, despite not being widely applied in practice, to solve the problem of increasing demand allocation in airports. The main objective is to address challenges not solved by the administrative mechanisms (BALL; DONOHUE; HOFFMAN, 2006).

The article Gillen, Jacquillat & Odoni (2016) cites two primary mechanisms for the economic management of the exposed problem: (i) Congestion pricing, where the cost of using the overloaded infrastructure is passed on to the air operators, which are interested to gain access to the airport; and (ii) auction of slots, as well as in the administrative management of demand, the airport declares the number of slots available through airport parameters, and it is given access to the air operator with the highest monetary offer.

Congestion pricing

The Figure 3.6 demonstrates the idea of price mechanisms, illustrating the relationship between the marginal cost of delay and the marginal benefit of allocation (GILLEN; JACQUILLAT; ODoni, 2016). Therefore, when the congestion price is not present, air operators can allocate “A” movements, resulting in a marginal delay cost “W”. However,

the total marginal cost, “Y”, exceeds the allocation benefit. Reducing the amount of the number of moves allocated to $B \leq A$, the total marginal cost is equal to the benefit of allocation “Z”, having an internal marginal cost “X”. There are, however, some difficulties with this demonstration, as (GILLEN; JACQUILLAT; ODONI, 2016) points out: (i) developing the congestion pricing scheme is complex, i.e., measuring the internal and total marginal cost is not trivial; and (ii) there are uncertainties regarding how the allocation of air operators varies with the congestion price.

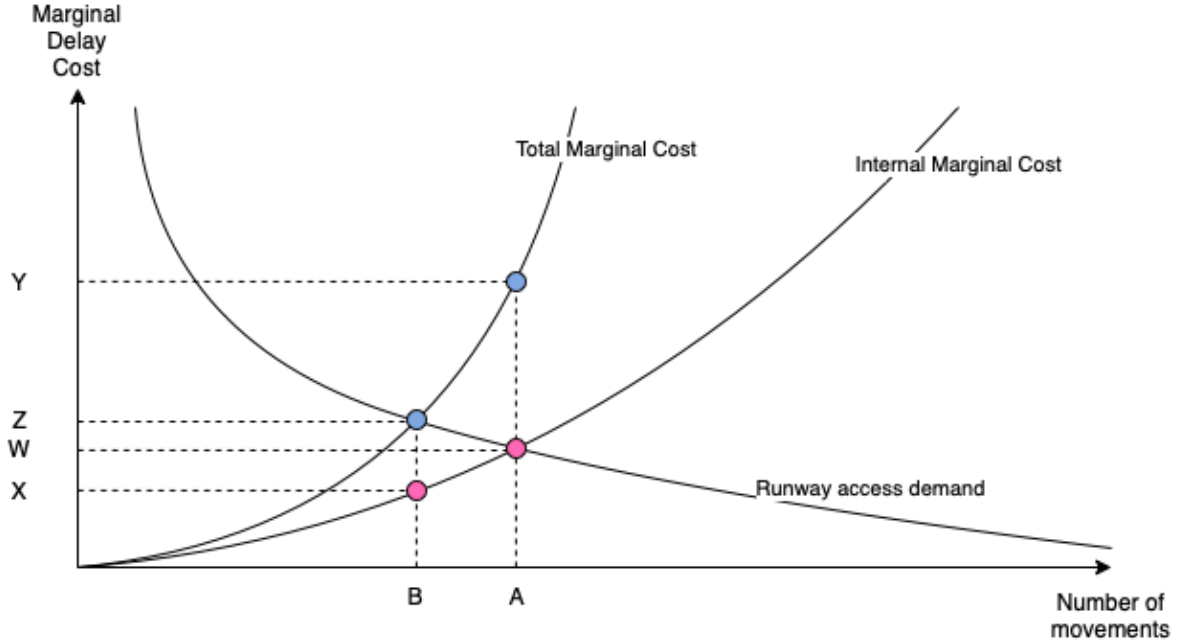


Figure 3.6: Congestion pricing as an alternative for external delay correction and slot allocation. Adapted from: (GILLEN; JACQUILLAT; ODONI, 2016).

Still about the economic management problem, it is important to point out that typically air operators allocate several movements in a short period, imposing congestion costs on themselves. It can be argued that airport systems containing hubs of few air operators, i.e., reduced competition, have a high internalization of congestion cost, reducing the price per slot, and making impacts related to economic demand management negligible (BALL; DONOHUE; HOFFMAN, 2006; GILLEN; JACQUILLAT; ODONI, 2016). However, for these airports, where the internal marginal cost is equal to the total marginal cost due to the lack of competition, it is shown higher rates of delays since overschedule are made by the monopolist air operator, that allocates demand which the operational cost cannot be totally internalized (BALL; DONOHUE; HOFFMAN, 2006). The Figure 3.7 demonstrates the relationship between competition and increased cost per slots. In a non-competitive airport system, the cost of delay is internalized and there are no

incentives to manage demand. However, introducing competition, it makes the cost per slots increase, distribution mechanisms that search for a balance between accepted moves and total delay cost are needed.

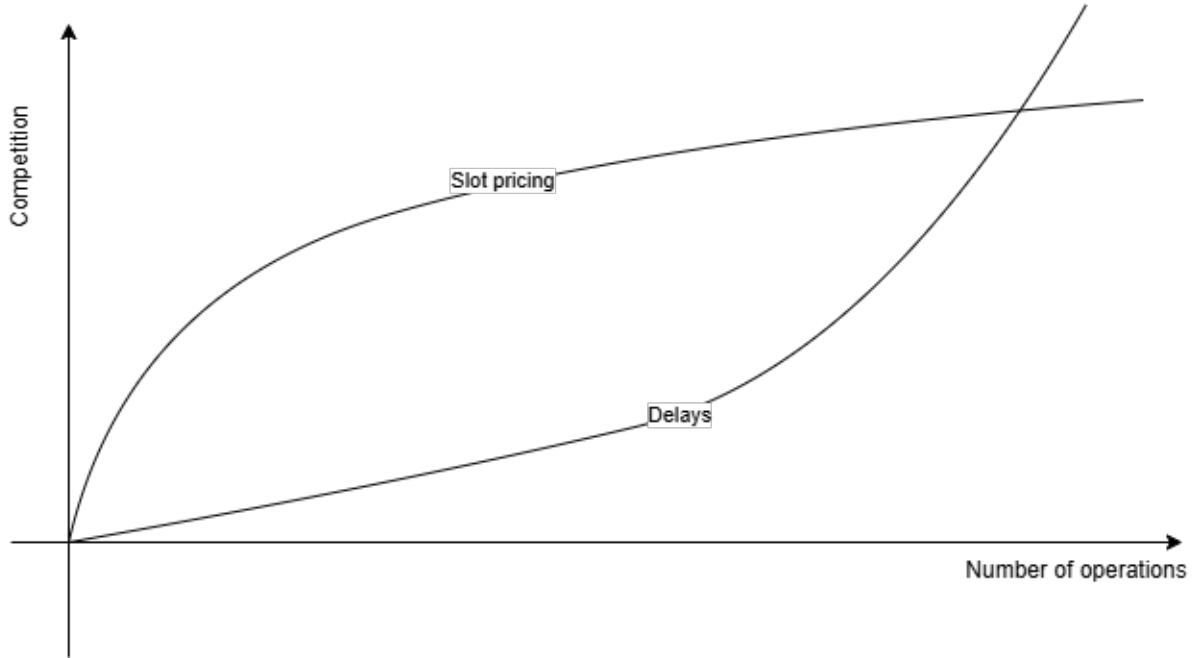


Figure 3.7: Relationship between the price of slots and costs related to competition. Adapted from: (BALL; DONOHUE; HOFFMAN, 2006).

Since congestion pricing is a topic that still requires further studies, due to its complexity in measuring the internalization of delays and designing the congestion pricing scheme for each modality of air operator (GILLEN; JACQUILLAT; ODoni, 2016), other economic management mechanisms would be the auctions of slots. These would also not be atomized, i.e., each slot auctioned individually, due to the risk of exposure to the air operator if arrivals and departures at a given airport do not end up coupled with each other and among other movements at other airports. This brings the need for combinatorial auctions, which need to be mutually exclusive.

Auction of slots

The knowledge that the slot will be auctioned in the future brings the continuous need to evaluate the market and the slots' price, limiting the negative sides of the administrative management of slots: (i) inefficiencies guaranteed by historic; (ii) legal difficulties in congested airports to appoint holders of newly released slots; and (iii) limitation of market entry for new air operators, as they have a less economic impact on the airport. It is

worth mentioning that the revenue generated by the auction is related to the market value due to the scarce system, paying for the cost of the used infrastructure. However, a slot auction system should focus primarily on encouraging competition, rather than maximizing revenue (BALL; DONOHUE; HOFFMAN, 2006).

For the auctions mechanism, (BALL; DONOHUE; HOFFMAN, 2006) points out limitations in 5 pillars:

- **Capacity:** the number of slots available for auction is related to demand, which is unknown before the auction takes place. Therefore, before declaring slot holders, it is necessary to ensure that the combination of slots granted does not exceed airport parameters.
- **Ownership:** it is necessary to specify the period of slots ownership, together with their acceptable operating range, which also influences the capacity. Regulation that allows secondary auctions, where ownership over the slot can be transferred between operators, is also needed. Daily transfers between operators may also be possible, allowing better use of airport infrastructure.
- **Value:** although the value of a slot is present to air operators, its value during an auction is unknown. Therefore, the market will evaluate and pay the slot holder an amount that will be corrected over time.
- **Market Power:** it is necessary to design auctions that have a regulation that encourages competition, preventing offers to buy a monopoly.
- **Implementation:** even if the transition from administrative management to an economic management system is organized in phases, i.e., auctioning a percentage of the slots in a congested airport, it will impact operators' connection networks airlines as a whole, and their losses to the market may be greater than the benefits.

A hybrid use is possible, i.e., combining the advantages of administrative management with economic management. Thus, the allocation could first come from an administrative allocation process (e.g., WASG and adaptations made by the local authorities), selling at congestion price or auctioning the remaining slots. This would add up to three important benefits of administrative mechanisms: (i) it helps air operators to project future decisions related to slots; (ii) it makes the process less complex by following the activities calendar, which proposes allocations twice a year; and (iii) it assists in the long-term industry projection, which can easily predict the growth of competition and the market, with the incentive of congestion pricing and slot appreciation, offered by economic management. However, it is worth mentioning that the disadvantages of both management would also be aggregated, where regulatory needs for slot distribution, ownership definition, and

market access would be evident, and could be disputed in court by airline operators, as occurred in 2008 in an attempt to auction 10% of New York airports' slots (CAVUSOGLU; MACARIO, 2021; GILLEN; JACQUILLAT; ODoni, 2016; BICHLER et al., 2021).

3.1.5 Airports free of demand management mechanisms

As well as economic demand management, the approach to airports with systems free of demand management mechanisms goes beyond the scope of this work, but it has been applied in American airports and its cost-benefits have been highlighted (CAVUSOGLU; MACARIO, 2021; GILLEN; JACQUILLAT; ODoni, 2016).

There are two objectives to be achieved at airports in terms of demand and slots: (i) maximize the number of movements accepted in the airport infrastructure, adapting required schedules to possible operating hours; and (ii) minimize congestion-related costs, which are present in high-demand airports in different forms (e.g., delays) (CAVUSOGLU; MACARIO, 2021). It is noted that the objectives are inversely proportional because as the number of movements in an airport increases, projecting growth in revenue, the delays and impacts for the air operators increase, as illustrated in the Figure 3.8 (ZOGRAFOS; MADAS; ANDROUTSOPOULOS, 2017; SWAROOP et al., 2012; GILLEN; JACQUILLAT; ODoni, 2016).

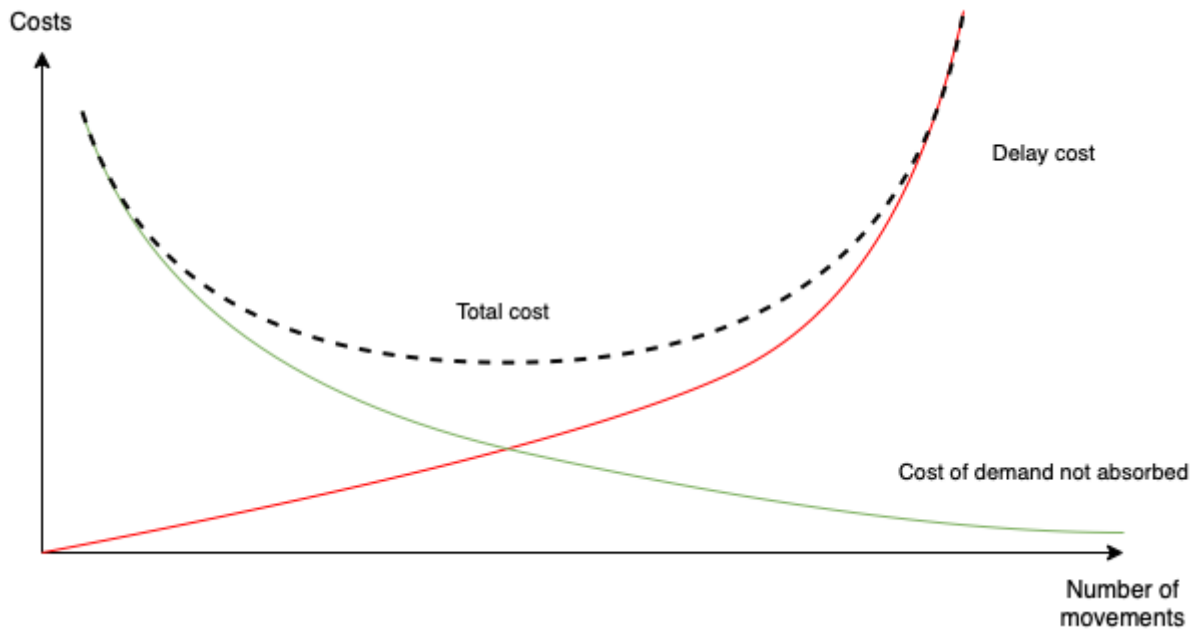


Figure 3.8: Cost-benefit of accepting movements in congested airports. Adapted from: (SWAROOP et al., 2012; GILLEN; JACQUILLAT; ODoni, 2016).

Although airports with high movement acceptance, and without demand management, are the simplest solution for allocation distribution, taking better advantage of the airport infrastructure (GILLEN; JACQUILLAT; ODoni, 2016), the approach also has risks and disadvantages.

The indiscriminate use of infrastructure helps generate revenue from stakeholders, who invest in the growth of the industry. In (ZOGRAFOS; MADAS; ANDROUTSOPOULOS, 2017) the authors argue that demand management often constrains the declared capacity of the airport, which in turn affects not only the slots accepted but the distance between the desired times of movements, i.e., the quality of slots in the airport in an air operator perspective. Thus, reducing management provides a greater frequency of movement, competition between operators, opportunities for new entrants, and better use of scarce infrastructure. On the other hand, the opening would also reflect on delay costs and slot reliability.

In (SWAROOP et al., 2012) the cost-benefit of applying demand management in airports with few mechanisms (e.g., LGA) was studied. It brings a methodology to simulate the cost related to increased movement and the costs generated by delays. Adding the benefits of 16 US airports in the simulation, the reduction in delay costs for passengers would be more than 200 million dollars annually. The study also proposes a reduction in infrastructure capacity, i.e., fewer movements during congested hours, resulting in a projection greater than 600 million dollars annually in benefits for passengers. However, the authors make it clear that, although the use of demand management mechanisms can generate financial benefits and a better level of service for passengers, by excluding movements, they could seriously harm the network of affected air operators. The costs generated with the implementation of these mechanisms were also not added. Although, intuitively, reducing delay costs also reduce operating costs, consequently the price of tickets for passengers.

Another important topic brought up by demand management mechanisms is capacity declaration. When an airport without these mechanisms is compared with coordinated airports, such as Frankfurt Airport (FRA) and Newark Liberty International Airport (EWR), which have the similar infrastructure, Figure 3.9, there is a concentration of movements in EWR, while FRA has a more balanced distribution. This characteristic is common in uncoordinated airports, where there are periods which scheduled movement exceeds the recommended limit of demand absorption, even in Instrumental Meteorological Conditions (IFR), i.e., adverse weather conditions of operation. In EWR, there is overutilization in the night period and underutilization in the rest of the day, showing inefficiency in the allocation. This factor affects the airport's punctuality performance, which, due to the non-linear relationship between the number of movements and delays

when the airport operates close to its capacity limit, makes certain American airports perform worse when compared to equivalents in Europe (GILLEN; JACQUILLAT; ODONI, 2016).

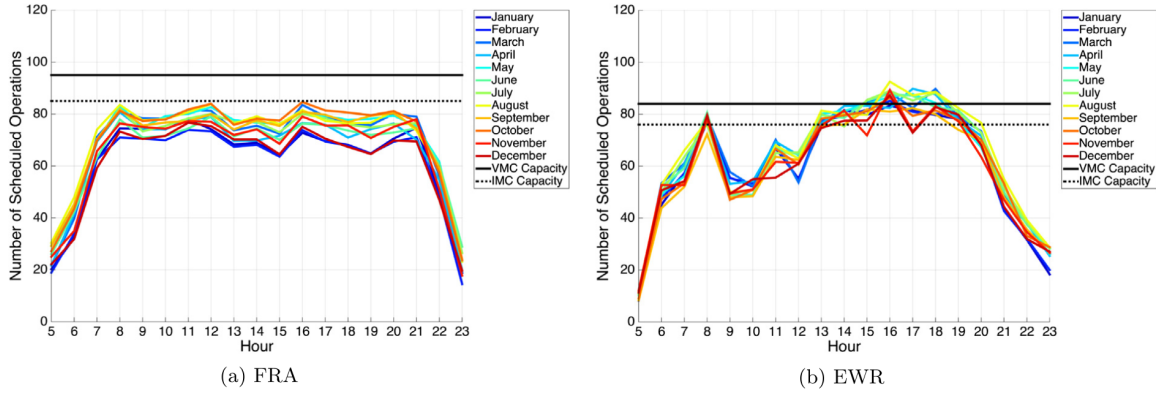


Figure 3.9: Average weekly movements per month for EWR and FRA airports in 2007. Source: (GILLEN; JACQUILLAT; ODONI, 2016).

The article (CAVUSOGLU; MACARIO, 2021) also points out that airports subject to coordination mechanisms tend to use the Visual Meteorological Conditions (VMC) operational limit of the runway as their maximum capacity, i.e., normal weather conditions. This is due to infrastructure overload due to adverse weather when delays are frequent. Operating above this conditional, as shown in the 3.9 part (b), even in VMC, already configures acceptance of excessive delays and low confidence in slot schedules, which would worsen in situations of IFR. In addition, the study (ZOGRAFOS; MADAS; ANDROUTSOPOULOS, 2017) points out that common concentrations in small time slots also increase the operational delay. By reducing infrastructure capacity, movements would be spread over new slots, minimizing the airport's overall delay. The Figure 3.10 illustrates the movement of the largest airport in Greece, where a reduction from 85% (part a) to 71% (part b) has a considerable impact on the airport's peak delays.

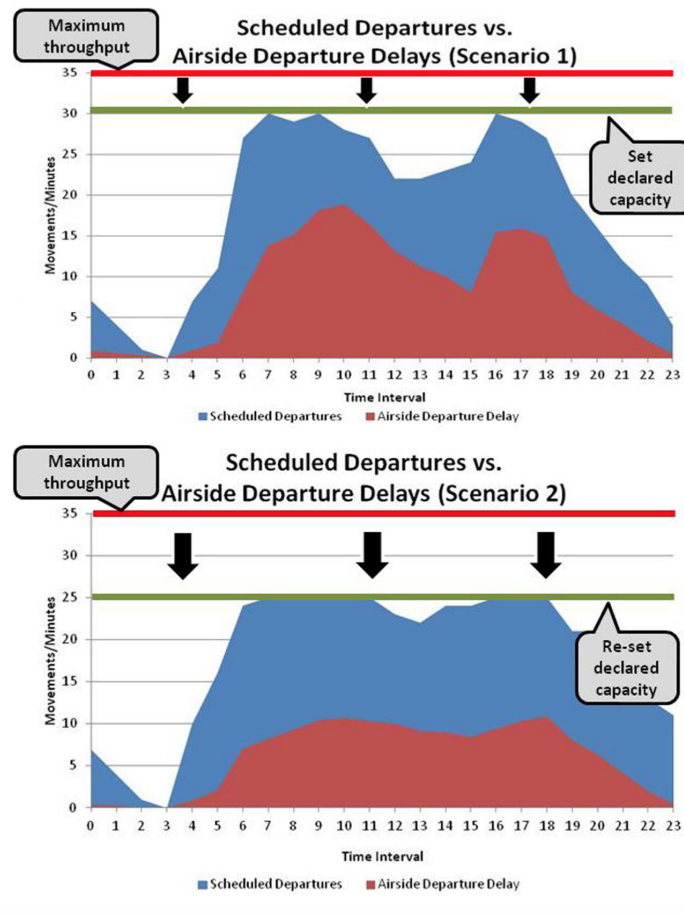


Figure 3.10: Modeling the effects of capacity declaration on congestion-related delays. Source: (ZOGRAFOS; MADAS; ANDROUTSOPOULOS, 2017).

Although there are difficulties in demand management systems (e.g., inefficiencies in the general use of infrastructure guaranteed by history, market opening, capacity declaration, and implementation) its mechanism provides a better definition of infrastructure parameters, which as consequence provides better demand absorption at adequate service levels (MORISSET; ODONI, 2011).

3.2 Queueing Theory

3.3 Model Checking

3.4 Conclusions

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